Three Elementary Inequalities and Their Geometric Meaning

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Cauchy–Schwarz, AM–GM and the triangle inequality appear in almost every early inequality exercise. Their proofs have different mechanisms: positivity of a square, concavity of logarithm, and the geometry of a norm. The presentation follows the standard inequality toolkit of Hardy, Littlewood and Pólya's *Inequalities*, with the vector interpretation aligned with MIT 18.06.[1]

§1.1 Squares give sharp inequalities and equality cases

For real a, b, nonnegativity of $(a-b)^2$ gives

$$a^2 + b^2 \geq 2ab,$$

with equality if and only if $a = b$. Adding this identity over pairs yields

$$2(a^2 + b^2 + c^2 - ab - bc - ca) = (a-b)^2 + (b-c)^2 + (c-a)^2 \geq 0.$$

Equality holds precisely at $a = b = c$. This pattern is called a sum-of-squares certificate: it proves the inequality and simultaneously explains when the deficit vanishes.

§1.2 Cauchy–Schwarz from the discriminant

For vectors $u, v \in \mathbb{R}^n$, the quadratic

$$q(t) = \|u - tv\|^2 = \|v\|^2 t^2 - 2\langle u, v\rangle t + \|u\|^2$$

is nonnegative for every real t. When $v \neq 0$, its discriminant must be nonpositive, so

$$\langle u, v\rangle^2 \leq \|u\|^2 \|v\|^2.$$

When $v = 0$, both sides vanish. Equality for nonzero v occurs exactly when $u = tv$ for some scalar t. The equality condition is a dependence relation, not merely a coincidence of numbers.

For $a_i > 0$, substitute

$$u_i = \frac{x_i}{\sqrt{a_i}}, \qquad v_i = \sqrt{a_i}.$$

Cauchy then becomes its Engel form:

$$\boxed{\sum_i \frac{x_i^2}{a_i} \geq \frac{(\sum_i x_i)^2}{\sum_i a_i}.}$$

Equality holds when x_i/a_i is independent of i.

[1] https://web.mit.edu/18.06/www/index.shtml

Example 1.2.1

For $a, b, c > 0$, applying Engel with denominators $b+c, c+a, a+b$ gives

$$\frac{a^2}{b+c} + \frac{b^2}{c+a} + \frac{c^2}{a+b} \geq \frac{(a+b+c)^2}{2(a+b+c)} = \frac{a+b+c}{2}.$$

Equality requires $a/(b+c) = b/(c+a) = c/(a+b)$, equivalent in this positive setting to $a = b = c$.

§1.3 The triangle inequality is the norm form of Cauchy

For Euclidean vectors,

$$\|u+v\|^2 = \|u\|^2 + 2\langle u, v\rangle + \|v\|^2 \leq (\|u\| + \|v\|)^2,$$

so $\|u+v\| \leq \|u\| + \|v\|$. In $\mathbb{R}$ this is $|a+b| \leq |a| + |b|$; substituting $u = a-b$, $v = b-c$ gives

$$|a-c| \leq |a-b| + |b-c|.$$

Rearranging the triangle inequality in both directions gives the reverse form

$$\big|\|u\| - \|v\|\big| \leq \|u-v\|.$$

In Euclidean space, equality in the forward triangle inequality occurs when the nonzero vectors point in the same direction. In a general normed space equality can also occur for other configurations; for example $\|(1,0)+(0,1)\|_1 = 2 = \|(1,0)\|_1 + \|(0,1)\|_1$. This shows why the equality criterion needs to mention the norm.

§1.4 AM–GM follows from concavity

For positive $x_1, \ldots, x_n$, the second derivative of the logarithm is

$$(\log x)'' = -x^{-2} < 0,$$

so log is strictly concave. Jensen's inequality gives

$$\log\left(\frac{x_1 + \cdots + x_n}{n}\right) \geq \frac{1}{n}\sum_i \log x_i = \log\left(\prod_i x_i\right)^{1/n}.$$

Exponentiation yields

$$\boxed{\frac{x_1 + \cdots + x_n}{n} \geq (x_1 \cdots x_n)^{1/n}.}$$

Strict concavity forces equality exactly when all x_i are equal. The formula extends to nonnegative variables by continuity; when any variable is zero, the right-hand side is zero, and equality requires all variables to be zero.

For $x > 0$, apply this to x and $1/x$:

$$x + \frac{1}{x} \geq 2,$$

with equality at $x = 1$. For $x, y, z > 0$, the product of x/y, y/z, z/x is 1, so

$$\frac{x}{y} + \frac{y}{z} + \frac{z}{x} \geq 3,$$

with equality at $x = y = z$.

§1.5 Coordinate geometry is an inner-product calculation

Let $A = (x_1, y_1)$, $B = (x_2, y_2)$. The distance formula is

$$|AB|^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2 = \langle B - A, B - A \rangle.$$

For direction vectors u, v, perpendicularity is the invariant relation $\langle u, v \rangle = 0$. When both lines have finite slopes m_1, m_2, direction vectors may be chosen as $(1, m_1)$, $(1, m_2)$, so perpendicularity becomes $1 + m_1 m_2 = 0$.

The slope rule $m_1 m_2 = -1$ does not cover vertical lines, but the dot-product rule does. Likewise, checking parallelism via a determinant $\det[u\ v] = 0$ works whether the slopes exist or not.

§1.6 A matrix version of the same square argument

For a real symmetric matrix A, the condition

$$x^T A x \geq 0 \qquad \text{for all } x$$

defines positive semidefiniteness, denoted $A \succeq 0$. For

$$A = \begin{pmatrix} a & b \\ b & c \end{pmatrix},$$

the exact criterion is

$$a \geq 0, \quad c \geq 0, \quad ac - b^2 \geq 0.$$

If $a > 0$, completing a square gives

$$ax^2 + 2bxy + cy^2 = a(x + (b/a)y)^2 + (c - b^2/a)y^2;$$

the case $a = 0$ requires $b = 0$ and $c \geq 0$. This is the two-dimensional analogue of proving an inequality by writing a sum of squares.

The three elementary methods now have distinct mechanisms: Cauchy is controlled by a nonnegative quadratic and linear dependence; AM–GM is controlled by concavity and equality of inputs; the triangle inequality describes metric geometry. Choosing the method means recognizing which mathematical structure the problem actually supplies.